Grybot: A Didactic Guitar Hero Robot Player on FPGA

Luiz Adriano A. O. Vaz and Marcus Vinicius Lamar
Dept. of Computer Science
University of Brasilia
Brasilia, Brazil
Email: l.diano.l@gmail.com lamar@unb.br

***Abstract*—The development of machines able to solve problems as well as human beings is one of the goals of robotics. Guitar Hero is a popular game in which the player rolls their motor skills by pressing buttons according to the rhythm of a chosen song. Due to the high difficulty level of the game does not exist yet a robotic system able to outperform the best human players. The development of the robot, called Grybot, is motivated by stimulating the formation of young computer scientists and engineers in a playful environment. Grybot is an automatic robot dedicated to play the Guitar Hero III game in a PlayStation 2 console. It uses only a FPGA (Field Programmable Gate Array) device to detect notes by processing the component video signal in real time and, through a solenoid driver circuit, physically to press the buttons of a DualShock 2 control. The robot gets correct hits between 95% and 100% on every song in the game playlist and at any level of difficulty. In particular, it reaches 698,622 points and 98% correct hit on the song "Through the Fires and Flames" on Expert mode without using the *Star Power*, which multiplies the score obtained by the player. This score places the Grybot in the 241^{st} position in the world ranking according to the ScoreHero site.**

***Keywords*-Robot; Guitar Hero; FPGA; Artificial Intelligence;**

I. Introduction

Digital games have been shown as the most popular and broad category of computer applications. The video game market is estimated at several billion dollars in revenues. Currently, the game development significantly boosts the advancement of technology in virtually all areas of computing, such as man-machine interface, artificial intelligence, computer graphics, modeling of physical and biological systems, computer architecture, software engineering, signal processing, etc. The game development is one of the main areas that attract young people to the exact sciences.

The Guitar Hero, released in 2005, was a big hit among teenagers and young people until very recently. This success is due, mainly, by proposing marriage between the feeling of playing in a rock band with a game of simple mechanics. The introduction of a new interface between man and video game through a device that imitates an electric guitar was one of the highlights of that game. Research groups at several universities took advantage of that success to motivate students to develop robots that simulate the movements and decisions made by a real player, encouraging them to make progress in the fields of robotics, image processing, artificial intelligence among others.

This work aims to propose a research methodology employed in Grybot robot, shown in Figure 1. This robot has been developed since 2011 as a platform for scientific development and encouragement for training of young scientists. The Young Talents Program for Science, launched in 2012 and funded by CAPES (*Coordination for the Improvement of Higher Education Personnel*), is a Brazilian Government program to encourage scientific initiation. It is intended to freshmen students at the undergraduate level and aims to early introduce them in scientific research [1]. In our institution, one of the projects that attract scholarship students of this program, in the fields of engineering and computing, is the work with this robot. Adolescents enter a university and already have the opportunity to work, in a scientific way, with a theme that they have great experience and interest.

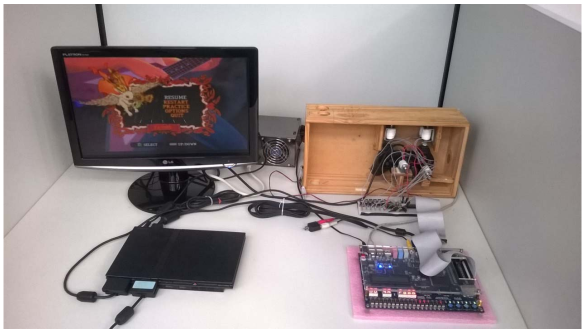

Figure 1: *Grybot robot*

The Grybot is a robot with a central processing unit based on only a FPGA (Field Programmable Gate Array). It has been designed with the goal of successfully play any music of Guitar Hero III - Legends of Rock (GH3) game. The GH3 was released in 2007 for the PlayStation 2 console and belongs to a music play game family, whose Guitar Hero and Guitar Hero 2 are its precursors. The signal analysis, image processing and control of the mechanical parts of the robot, are implemented directly in hardware using Verilog [2] hardware description language. This methodology avoids the use of a computer as a control agent, and allows the research and development of adaptive and evolutionary digital systems in a highly interesting and motivating environment for students.

Guitar Hero is a music play game first developed by Harmonix for the PlayStation 2 (PS2) and Xbox 360 consoles,

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which became very popular between 2005 and 2010 [3]. The player simulates playing an electric guitar using the standard controller of console or a special controller with the format of the musical instrument and similar mechanics.

The game screen shows the arm of a guitar as Figure 2. The musical notes, corresponding to colored disks, continually scroll down in accordance with the rhythm and melody of the song. The player's goal is to correctly play the music by pressing the control buttons at the exact moment that the disk crosses a specific region of the fretboard.

Figure 2: *Guitar Hero III game screen*

The game Guitar Hero has 6 official sequences: Guitar Hero II, Guitar Hero Encore: Rocks the 80s, Guitar Hero III: Legends of Rock, Guitar Hero: On Tour, Guitar Hero: Aerosmith, Guitar Hero World Tour. In addition to official versions, hundreds of customized versions were developed based on the engines of the original games. The game Guitar Hero III: Legends of Rocks (GH3), was chosen to be used in this work due it to be the most mature and popular among official versions. GH3 has a repertoire of 45 songs in the regular game and 25 bonus songs. As in other versions of the game, the speed, quantity and complexity of combinations of notes vary according to the chosen musical track and the difficulty level.

II. Related Work

The success of the game Guitar Hero has led several research groups, especially the field of robotics, take advantage of the interest of their students and to invest in researches on robots seeking the solution of this game. This section shows the result of an exhaustive search on projects of Guitar Hero automation developed by the scientific community until 2013. Due to the difficulty of obtaining technical data, the intention is not to go deep in the explanation and comparison of technical details of each implementation, but expose the state of the technology of similar robots developed until that year, to the best of our knowledge.

- **HeroNoid**

Following the fever that came with the launching of the game Guitar Hero in 2007, Mizrahi and Chalozin [4] developed the robot Heronoid, shown in Figure 3.

Figure 3: *HeroNoid [4]*

According to the authors its operation is based on two main components, the "brain" and the "body". The brain is responsible for the note detection, analysis of video signals and generating commands to the body. The body receives the commands and controls the movement of the "fingers". The commands generated by brain are sent via TCP (Transmission Control Protocol) to the body, and then translated to an appropriate signal and sent through a parallel port to the main control board of the body. This board receives the signal from the computer and drives a set of solenoids for pressing the buttons of the game controller.

During the game the color of a note is brighter than the colors of the remainder image. Starting from this premise, the authors define small trapezoidal regions for notes detection and represent the pixels of these regions in HSV (Hue, Saturation, Value) color space. A threshold is set to the brightness value thus obtained a binary representation of the pixels. If the region contains a certain amount of activated pixels then a valid note is detected.

The composite video output of the console is connected to a video capture card on computer. The device captures video frames and sends them to the image processing program that detects the notes. The detection signals related to notes for each color are then sent to a FIFO (First In First Out). After a while the removal of the note occurs and a command is sent to the control board of the guitar. Setting the time synchronization is accomplished manually according to the beats of the music. The authors say that this issue would be addressed in a future work.

- **Slashbot**

In 2008, it was built the robot SlashBot by Dave Buckner, Mitchell Jefferis, Vinny LaPenna, and Michael Voth from Texas A&M University [5]. Figure 4 shows a picture of the robot developed by the group.

This robot is also divided into two components. The "brain", the central processing unit comprises a system PXI^{TM} and a code written in LabVIEW ^{TM} which analyzes the video signal. The "body" is composed of the drive circuits and solenoid used to press the buttons on the game controller.

The system consists of three main modules: real-world

Figure 4: *SlashBot [5]*

interface, sensing & computing unit actuation component. The interface to the real world is responsible for producing the video signal and the interaction with the game. The sensing & detection unit is composed by a computing system designed to convert the analog video signal to a digital input. Also, this unit analyzes the input signal and sends control commands to the action component. The action component consists of seven solenoids. Five of them are positioned on the note action buttons, one is positioned to drive the strum bar, and the last one is positioned to press the "select" button, which triggers the *Star Power*. The notes on the screen are brighter compared to other objects in the scene. The brightness of a pixel is given by a analog signal from 0 V (black) to 1 V (white). The grey scale image analysis identifies the note disks with ease. Then, only the signs of some pixels, carefully chosen, are monitored using *NI LabVIEW*. When a note crosses through a monitored pixel there is a peak voltage in the intensity signal, indicating that the corresponding button should be pressed by the robot. Once a detection occurs, a signal enters in a queue. After a delay time, determined by the rhythm of the music, trigger signals are sent to the solenoids.

• **CythBot**

In 2008 *Cyth Systems* built the robot named CythBot [6] shown in Figure 5.

Figure 5: *CythBot [6]*

The Cythbot was designed using high-speed hardware and advanced technologies used in industrial process automation and product testing. The robot is composed by a collection of integrated components that detect the notes on the game screen and press the controller buttons by pneumatic actuators. According to the authors, the structure of the robot is represented by a producer-consumer model, in which the producer is a video camera, and the consumer is the guitar.

• **eGOR**

eGOR [7] (Electronic God of Rock) is the name of the robot created in 2009 by Kyle Harrison, Jacques Mainville, and Mike McGuire, shown in the Figure 6.

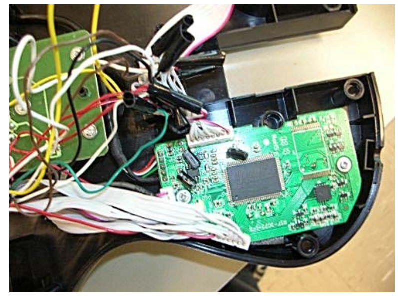

Figure 6: *eGOR [7]*

In this robot the image of the TV screen is captured by a camera. The video signal is sent to a *NI Vision Assistant* where it is filtered. The *NI Vision Builder* receives the filtered signal, detects the objects of interest and sends commands to a platform *NI USB-6008*. A 5 V signal is applied to a relay which control the signal that is transferred to the guitar. The process is applied to press the 5 buttons corresponding to the notes, to move the strum bar, and to trigger the *Star Power*. The camera acquires grayscale images at a frame rate of 30fps. The frames are sent to the *NI Vision Assistant* for filtering and noise reduction. The processing is applied only in the regions of interest defined by detection boxes placed on the points where the notes pass through. The sizes of boxes must be precisely calibrated, because affect significantly the performance in capturing fast sequences of notes. A detection box is also positioned at the *Star Power* meter to determine when to activate it.

• **DeepNote™**

In 2008, Jeremy Blum, Brandon Fischer, Zach Lynn, Alex Miller and Ben Shaffer developed a GH player robot named DeepNote ™ [8] shown in Figure 7.

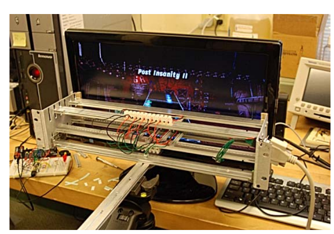

Figure 7: *DeepNote Guitar Hero Bot [8]*

The robot consists of a detection unit and activation circuits. A set of photodiodes is carefully positioned on each fret of the guitar on TV game screen. The aperture angle of capture of light by a photodiode is narrow and it has very short response time. These characteristics make photodiodes devices appropriate to accurately detect the light of a given pixel on screen. Rack and guitar are equipped with D-sub 15 pin connectors. Five pins are used to transmit signals from the sensors, one pin is used to transmit additional sync data and two are used to provide power. The “brain” of the robot is built with the *Parallax Propeller Propstick*, an eight-core microcontroller with 80 MHz clock frequency. The data are processed by a program written in the high-level language Spin. Digital signals generated by the rack are sent to the chip *Parallax* incorporated into the guitar circuit. The microcontroller has digital output pins used to provide electrical signals that directly trigger circuits guitar, without using any electromechanical devices.

- **AGH1000**

AGH1000 is the name given to the robot created in 2008 by Michael Seedman, shown in Figure 8 [9].

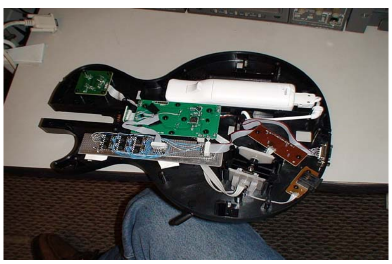

Figure 8: *AGH1000 robot [9]*

The system consists of a Nintendo Wii console, a TV and four processing boards. The Analog Processing Board is responsible for processing the composite video signal and generate digital signals corresponding to the notes detection. The digital processing board is powered by a Xilinx Spartan 3 FPGA device, configured by VHDL language, that processes the input digital signals and generates the control signals sent to the Driver Board. The Driver Board is composed of a single octal-buffer chip which controls the optical isolator board. This board directly commands the electronic circuits of the guitar. Also in this robot no electromechanical devices to press buttons are used.

- **Titan**

Titan, shown in Figure 9, is an automatic GH player created in 2009 by the FRC Team Titanium [10]. This robot uses an USB webcam to acquire the images. The processing is performed by a notebook that analyzes images and generates signals to drive the solenoids used to press the buttons on the guitar shape controller.

- **Others**

In 2009, Chee Lor and Mike Teigen, students from Fox Valley Technical College (FVTC) developed a GH robotic player [11]. In the same year, Pete Nikrin [12] proposed the Roxanne robot, based on vision system and a Programmable Logic Controller.

Figure 9: *Titan [10]*

The latest robotic GH player found by this research is named Guitar Zero created in 2012 by Eric Falk [13]. This robot was built to learn how to play a song using a supervised learning model. The player takes the role of teacher and manually plays the song once. The robot memorizes the sequence of commands and it becomes able to execute that song autonomously. Many people do not consider it a real robot, since it is just an electronic circuit installed inside the game controller using no sensors nor actuators.

It is interesting to note that all previously described robots use the specific Guitar shaped controller to interact to the PS2 game console. This work also aims to investigate an alternative approach by using a common DualShock2 game controller. The proposed approach allows Grybot to play not only GH based games, but any game of the PS2 console, just modifying the control circuit implemented in the FPGA.

III. Signal Processing with FPGA

Field Programmable Gate Array (FPGA) are configurable digital logic chips that allow the synthesis of any logic function [14]. An FPGA consists of thousands logic elements (LE). Generally, a LE is composed by a look up table (LUT), flip-flops, multiplexers, and other simple digital circuits. LEs are connected to each other through interconnection resources existing on FPGA. In general, systems implemented in FPGAs are designed from schematics circuits or coded using a hardware description language (HDL). The approach using schematic design can provide quick understanding of whole project. However, it becomes difficult to maintain large projects. VHDL and Verilog [2] are the most popular HDLs and the main languages supported by FPGA vendors. For a circuit to be implemented on a FPGA device, you must perform the steps of synthesis, placement and routing using a tool provided by the chip manufacturer. The synthesis process translates the design described in HDL or schematic into a netlist. The placement and routing are the processes in which the elements of the netlist are physically allocated and mapped to physical resources available in the FPGA.

This project has been implemented on a DE2-35 development kit, shown by Figure 10, manufactured by Terasic, using the Quartus-II design tool.

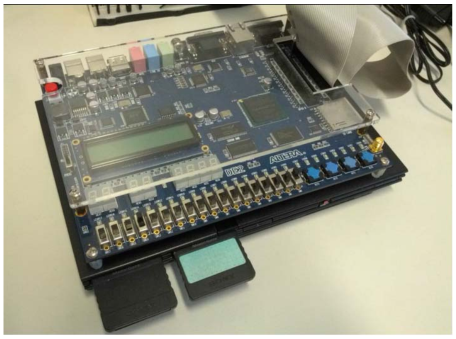

Figure 10: *DE2-35 Kit on top of PlayStation 2*

This kit contains several I/O interfaces connected directly to the pins of an Altera FPGA chip, Cyclone-II family, model EP2C35, existing on the board. The DE2-35 development kit include a VGA triple 10-bit high-speed Analog-Digital Converter (ADV7123) with VGA-out connector. Digital synchronization and color signals are generated by FPGA device, which implements the VGA controller. The ADV7123 chip is responsible for analog RGB output signals generation. This project uses the VGA signal with 480 rows, 640 columns, 60 frames/sec. Through this interface, the composite video signal generated by PS2 console is displayed on a VGA monitor. The DE2-TV Module, described in Verilog, is provided by the DE2-35 kit manufacturer. It receives the video composite signal input and generates the VGA signal output. As shown in Figure 11, this module consists of two major blocks, I2C_AV_Config and TV_to_VGA. The TV Decoder (ADV7181) and VGA DAC (ADV7123) chips control the signals input and output.

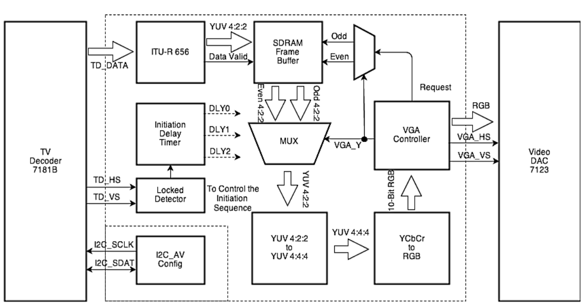

Figure 11: *DE2-TV block diagram*

After the initialization of the FPGA, the 'TV Decoder' chip configures the decoder through the block 'I2C_AV_config', which uses I2C communication protocol. The block 'ITU-R 656 Decoder' extracts the video signal, in YCrCb 4:2:2 format, from data stream and generates a control signal that indicates a valid data. The VGA controller sends an even/odd request to the 'SDRAM Frame Buffer' which demultiplexes odd and even fields from the interlaced composite video signal. The signal is converted from YCrCb 4:2:2 to YCrCb 4:4:4 format, and then to RGB format and finally delivered to the VGA controller. In this project, the SDRAM address is monitored to determine the (x, y) coordinates of current pixel on the screen.

IV. Proposed Robot Model

While a person plays a game, different information are automatically processed in his brain. The image is captured by the eyes. Next, the brain analyzes this image and decides the actions to be taken. It defines the time moment that the actions must occur, and sends commands to the body parts that perform them. The body reacts to the stimuli and the brain analyzes the effects of the actions, restarting the process.

A. *Design Decisions*

Similar to what happens to the human being, the sequence of actions should also occur in a robot. In this section, some techniques for implementing these processes are presented.

1) Visual information capturing: The first step to perform the actions required by the game is to capture the visual information. The robot needs a way to identify objects in the game. The most common way is the analysis of images obtained by cameras; an alternative technique is the capture and processing of composite video signal.

Camera: The wide variety of existing cameras, allows the designer to choose the type of sensor that best fits your needs and budget of the project. The computer vision techniques developed in the project can be easily applied in other studies since the implementation is not restricted to a specific video standard. On the other hand, real-time applications require fast processing of the video signal, and usually require a high computational effort. Thus, the cost / benefit of this approach could become decompensating or even unfeasible. Moreover, the images acquired by cameras suffer strong influence of the ambient light, reflections and distortions that may significantly affect the image analysis.

Composite video signal: The use of CVS does not require an image capturing device, like a camera. The CVS is available at the output of the video game console and can be directly processed by the robot. Besides, this approach is not affected by the characteristics of the environment in which the robot is located. However, CVS is an analog signal, then it is necessary the robot be equipped with a device for converting it into a digital signal. Furthermore, the processing of specific image regions becomes more complex, since the access to the pixel color information occurs sequentially, according to the signal scanning.

2) Notes detection: After setting how the video signal is captured, it is necessary to define a method to be used for notes detection.

Color: In the game, each note has a different color. The color-based approach uses the value of the component of RGB pixels to detect the exact time when the discs pass through the region of interest (ROI). In this approach the detection of notes becomes more specialized and only identifying objects with specific colors and disregarding possible identifications of objects from the background. However, when the user triggers the *Star Power* the notes change their colors, being all blue. Thus you need to take this issue if this feature is used during game play.

Shape: In the Guitar Hero game every musical note is represented by a colored disk or a rotating blue star, depending on the activation state of *Star Power*. The shape analysis of the note allows to differentiate common notes from special star notes. Furthermore, in the capturing stage it is not need to acquire the color of the notes. Moreover, the analysis of a image region is needed to carry out the shape recognition, requiring a more complex algorithm than a color detection algorithm.

Position: In general, computer vision systems use specific areas of the image to detect objects. In the game, the fretboard is shown in a linear perspective to give to the player the feeling that the notes come out from the display. Consequently, notes near the top of the screen are tiny, and at bottom of screen the notes have a larger number of pixels. For note detection it would be interesting that the method chosen could monitor the bottom part of the screen, where the notes are large and easily detectable. However, when a note is correctly played, an animation of flames arises from bottom to up, as show the orange note on Figure 2. This animation critically interferes with the detection of the next notes, making impossible the use of this region of the screen.

3) Time synchronization: Before the game starts, the player must select the difficulty level among Easy, Medium, Hard and Expert. The amount of notes of the music and their falling speed vary according to the chosen level. The robot must synchronize the time to press the buttons with the speed of the notes. So, it is necessary to develop a method to estimate the time required for a note to cross the whole screen. The measured time is used to set the parameters for timing and synchronization.

4) Game interaction: The robotic system must have capabilities to interact with the game. This can be accomplished by mechanical or electronic commands.

Electronic commands: The transmission of the command to the game system is accomplished through wiring the robot circuit directly to the video game controller circuit. This approach provides a reduced response time, low influence of the physical environment, and no problems related to the physical stress of the components. However, the wiring requires modifications of the internal components of the gaming system. For some developers this is considered a *hack*, it is seen as cheating, once directly operates the controller circuit.

Mechanical commands: Typically, the transmission of commands to the game is performed by pressing the buttons on the controller. Pneumatic or electrical (motors, solenoids, etc.) actuators can be used to execute the task. This approach performs a similar action to human fingers and it is non-intrusive to the console electronic system. However, it requires a precise positioning of the moving devices; it is influenced by the environment (temperature, humidity, etc.); and it suffers from mechanical wear along time.

B. The Grybot

The didactic robot Grybot, from initials of the note colors Green, Red, Yellow, Blue, and Orange, consists of a group of electromechanical devices that works together to automatically play games. Initially, the Guitar Hero 3 game is focused due to its simple gameplay. It is expected that more complex games would be solved by this robot in a near future. Figure 12 shows the diagram of the proposed robotic system.

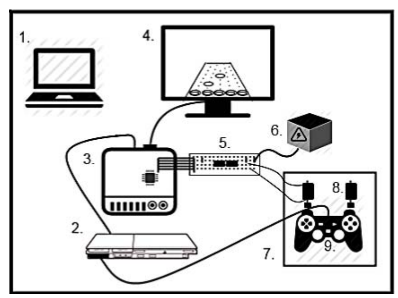

Figure 12: *Diagram of robotic system*

The main structural components of the system are:
1. PC computer
2. PlayStation 2 console,
3. Development Kit DE2-35
4. VGA monitor
5. Driver circuit
6. Power supply
7. Mechanical support
8. Electromechanical solenoids
9. DualShock 2 controller

Figure 13 shows the block diagram of the Grybot control circuitry implemented on FPGA.

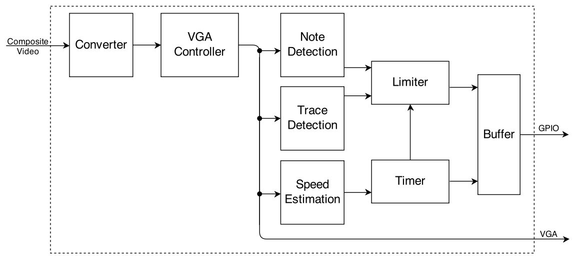

Figure 13: *Grybot control block diagram*

The PS2 console delivers a CVS to the Development Kit by a RCA connector. The CVS is decoded by an ADV7181 TV decoder chip into YCrCb video format and sent to the DE2-TV module, implemented on the FPGA chip. This module converts the YCrCb video signal into VGA output compatible signals in order to display the console screen in a computer monitor. The DE2-TV is the main module for acquiring and displaying video signals. From the VGA controller the synchronising signals, the position (x, y) on the screen, and the RGB values for the current pixel are extracted. These signals are sent to the following modules 'Note Detection', 'Trace Detection', and 'Speed Estimation'. The outputs of these modules control the 'Limiter', 'Timer' and 'Buffer' modules. The commands for the electromechanical devices controller are sending by the GPIO (General Purpose Input Output) interface of the Development Kit. Technical details of each module are presented in the following sections.

1) Detection System: The detection system analyzes image by scanning VGA output signal. This system is divided into 'single-note detection', 'continuous-note detection', and 'speed estimation' modules. Each module examines a specific group of pixels according to its function.

Single-note detection module:

This module detects single-notes by monitoring a group of 5 pixels shown in Figure 14. One pixel is positioned on each fret of the guitar. To avoid interference, the positions of these pixels are set above the flames animation that arises in a note hit.

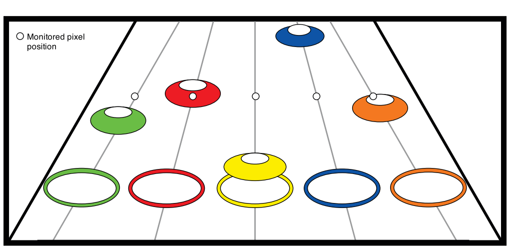

Figure 14: *Single-note detection regions*

When the scanning VGA signal reaches a monitored pixel position, the pixel color (R_i, G_i, B_i) is analyzed generating a D_i signal indicating detection of valid single note. The color of the pixel is compared with a previously defined color template for each fret by

$$D_i = \begin{cases} 1, & (R_{\min} < R_i < R_{\max})\ and \\ & (G_{\min} < G_i < G_{\max})\ and \\ & (B_{\min} < B_i < B_{\max}) \\ 0, & otherwise \end{cases} \tag{1}$$

where $(R_i, B_i G_i)$ is the color of current pixel, $(R_{min}, G_{min}, B_{min})$ is the lower limit and $(R_{max}, G_{max}, B_{max})$ the upper limit defining a detection cube in RGB color space.

Continuous-note detection module:

The trace detection module monitors 5 sets of pixels, shown in Figure 15 by rectangles. Each set of pixel is composed by 13 pixels positioned in line.

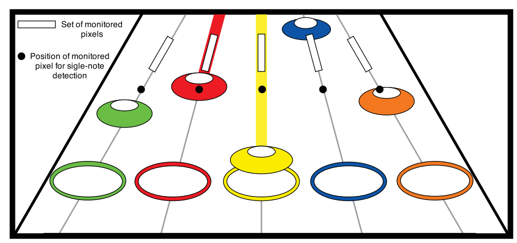

Figure 15: *Continuous-note detection regions*

The presence of a continuous-note C_i on fret i is given by

$$C_i = \begin{cases} 1, & \sum_{k=1}^{13} D_i(k) \geq 11 \\ 0, & otherwise \end{cases} \tag{2}$$

where $D_i(k)$ is detection of pixel k from Equation 1.

Speed estimation module:

The speed of notes in a music is estimated by analysing 2 points on a fret shown in Figure 16.

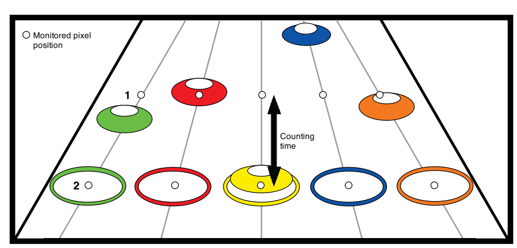

Figure 16: *Speed estimation regions*

The speed of the notes is estimated by the time needed to a note to displace from point 1 to point 2 of a fret. A 27MHz clock signal is used to drive a counter which starts when a valid note is detected at point 1, and stops when it reaches point 2. This speed depends on chosen music and selected difficulty level. However, the speed does not change during music playing .

2) Frequency Limiter: In the game, several songs have passages which a note is quickly repeated, as shown in Figure 17. The correct triggering of solenoids is difficult due to the small distance between these notes.

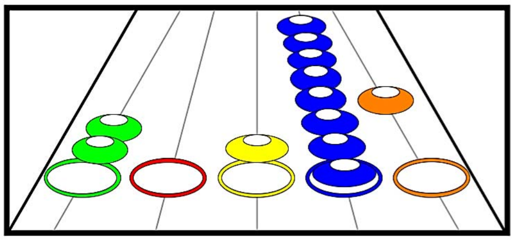

Figure 17: *Fast sequence of a note*

To minimize this issue a Limiter module has been included. The maximum frequency of control signals is limited by this module. As any mechanical device, electromechanical solenoids can only respond efficiently to commands applied by a minimum period of time. This time is necessary for coil be energized and magnetic force overcomes friction and plunger inertia, then moving it to the final position. Similarly, the solenoid disengagement also requires a minimum period of time, which is needed by return spring to move plunger to its initial position. These times are experimentally calibrated according to mechanical and electrical characteristics of the used solenoids.

Delay Buffer:

The screen region used for note detection is different from region which the notes must be played by robot. Thus, a time delay is needed. A buffer is used to delay note detected signals. The amount of delay is calculated from the estimated speed. The delayed signals are then forwarded to solenoid driver circuits through the GPIO interface.

Timing:

In the GH3, the player may chose one of 4 difficulty levels and, in training mode, one of 4 note speeds. Therefore, a song can be played at 16 different ways. The timing module consists of a Look Up Table (LUT) that non-linearly maps the estimated music speed in values for Limiter and Buffer modules. The LUT consists of 16 speed intervals which classify the input estimated speed in counter values for the next modules. The non-linear mapping of speed allows the Grybot obtains an accurate delay time for each level and speed combination.

3) Actuation system : The actuation system receives electrical control signals from FPGA through GPIO interface and converts them into mechanical movements by solenoids. Solenoids are electromechanics devices composed by a coil and a magnetically moved plunger. The device requires high current to energize the coil in order to perform the plunger movements. When sufficient current is applied to coil plunger is pulled by magnetic force. When the current ceases plunger is pulled to original position by a spring. The GPIO interface is not able to supply required current. Thus, the electronic circuit, shown in Figure 18, is built to drive each solenoid. Optocouplers are used to isolate low level 3.3V control circuits from 12V actioning circuits.

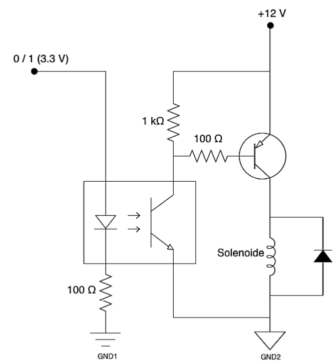

Figure 18: *Solenoid driver circuit*

The transistors are used as solid state switches to reduce the actuator response time. Solenoids are positioned so that their movements can press the DualShock 2 controller buttons, as shown in Figure 19.

Figure 19: *Solenoids positioning*

V. Obtained Results

The control system is implemented on a Altera EP2C35 FPGA chip. This chip contains 33,216 logic elements, 483,840 RAM bits, 35 embedded 18 bits multipliers, 4 PLLs (Phase Locked Loop) and 475 user I/O pins. Detailed information about the resources used in the design is shown by Figure 20.

Though this design is quite complex, it uses only 43% of the FPGA LE resources. Thus, there are still remain available resources to make advances in main control module.

Flow Status	Successful - Mon Sep 30 23:41:26 2013
Quartus II 64-Bit Version	12.1 Build 243 01/31/2013 SP 1 SJ Web Edition
Revision Name	DE2_TV
Top-level Entity Name	DE2_TV
Family	Cyclone II
Device	EP2C35F672C6
Timing Models	Final
Total logic elements	14,283 / 33,216 (43 %)
Total combinational functions	13,946 / 33,216 (42 %)
Dedicated logic registers	4,082 / 33,216 (12 %)
Total registers	4082
Total pins	426 / 475 (90 %)
Total virtual pins	0
Total memory bits	53,184 / 483,840 (11 %)
Embedded Multiplier 9-bit elements	44 / 70 (63 %)
Total PLLs	1 / 4 (25 %)

Figure 20: *FPGA Flow Summary*

A. Performance

Initially, user must select game mode, music, and difficulty level, once the Grybot is yet not able to automatically set them from the first menu screen. In GH3 game, performance achieved by player is presented at end of each song played. The total points (*Score*), maximum number of consecutive correct played notes (*Streak*), and percentage of properly hit notes (*Hits*) are displayed on the screen. Videos of the Grybot can be watched in [15].

Through The Fire And Flames (TTFAF) is considered the most difficult GH3 music. This song represents a great challenge to the player, because of its long duration, high speed notes, and many different combinations of notes. Assuming that TFFAF is also the worst case for a robot, it is used to present detailed results. Apart from selectable difficulty levels, the game also has a training mode where the errors made by the player do not cause the end of the game. For each level, the player can practice any song selecting one of 4 different speeds: Slow, Slower, Slowest and Full Speed. The averages of obtained scores, streaks, and hits, for each difficulty level and speed are presented by Table I.

Table I: *Obtained performance for the TFFAF song in training mode*

Level - Speed	Score	Streak	Hits
Easy - Full Speed	213,281.0	649.0	100%
Easy - Slow	211,961.8	649.0	100%
Easy - Slower	211,780.8	649.0	100%
Easy - Slowest	212,936.6	649.0	99%
Medium - Full Speed	383,412.4	641.0	100%
Medium - Slow	384,619.4	644.6	100%
Medium - Slower	383,610.0	641.0	100%
Medium - Slowest	381,027.2	641.6	100%
Hard - Full Speed	455,939.2	621.4	97%
Hard - Slow	523,695.6	775.4	98%
Hard - Slower	459,030.6	359.2	99%
Hard - Slowest	483,381.8	434.4	99%
Expert - Full Speed	698,622.6	1,118.2	98%
Expert - Slow	739,835.6	662.4	99%
Expert - Slower	708,992.6	469.2	99%
Expert - Slowest	746,784.6	1,066.4	99%

At Full Speed & Expert level the robot reaches a 98% of correct note hit. As the robot is calibrated to play the worst case, in Full Speed & Hard level its performance its a little bit worse, achieving 97% correct hit. The errors are mainly due to failures of the note detection modules or, even if notes are correctly detected, the solenoid fails to properly press a controller button. In [16] is presented the complete table with results for every music and speed of the game.

B. Comparison with other robots

Grybot is the only robot that uses conventional DualShock2 controller to play the GH3 game. All other robots found in this study use a guitar shaped controller. The Grybot is supposed to be a first prototype of a robotic general game player. Therefore, it is designed to use a generic controller in order to be able to play another games in a near future.

Table II presents a comparison between Grybot and other reported results for single player, Quick Play (full speed) mode and Expert difficulty level, for the songs *Cliffs of Dover* (COD), *My Name is Jonas* (MNIJ), *Cult of Personality*(COP) and *Through The Fire And Flames* (TTFAF).

The Slashbot uses CVS signal to analyze image pixels, and send signals to drive a solenoid based electromechanical system. Unlike Slashbot, Grybot does not have a driver for pressing the button that triggers the *Star Power*. When *Star Power* is activated awarded points are multiplied by 2, 4, or 8, according "Power" intensity. The authors of Slashbot report problems concerning automatic estimation of music speed, thus the delay time is set manually for each music. Also, some difficulties to distinguish continuous-note and rapid sequences of notes are reported [5], resulting in a lower score in TTFAF music. The Titan robot is also able to use *Star Power*. In despite that, Grybot reaches a higher score. DeepNote robot shows better performance compared to Grybot. The difference between the percentage of correct hits is 1 % in music TFFAF. The difference between the scores is 66,016 and 176,642 respectively. In the COP song, the AGH1000 score is higher than that achieved by Grybot. However it was only possible due to the activation of *Star Power* have been carried out by an human agent during AGH1000's game play, as shown in [17]. The robot AGH1000 failed to successfully play the hardest song TTFAF.

C. Grybot vs Human beings

The comparison of performance of the robot Grybot and human players is made based on results registered on the collaborative ScoreHero site [18]. This platform was created in January 2006, and allows GH players submit attained scores on various songs and difficulty levels. To provide reliability to received data, players must submit evidences of the achievements, as screen shots showing the results or videos shared by YouTube. According to the site ScoreHero,

Table II: *Grybot vs Other Robots*

	Grybot		Slashbot		Cythbot		Titan		eGOR		DeepNote		AGH1000	
	Score	Hits	Score	Hits	Score	Hits	Score	Hits	Score	Hits	Score	Hits	Score	Hits
COF	252,656	99%	122,761	96%	-	99 %	246,355	98%	-	-%	-	-%	225,123	99%
MNIJ	212,220	98%	169,432	97%	-	-%	-	-%	-	-%	-	-%	-	-%
COP	246,458	99%	-	- %	-	99 %	-	-%	246,304	98%	312,474	99%	-	-%
TTFAF	698,622	98%	-	- %	-	82 %	-	-%	485,757	91%	846,540	99%	-	-%

the Grybot robot got the 241^{st} rank position, with 698,622 points and 98% accuracy on the TFFAF song in Expert mode. The first place taken by the player Altruit reached 984,744 points and 99% accuracy. The difference between the scores are relatively large compared to the difference of the percentage of correct hits, due to the fact that Grybot can not trigger the *Star Power* mechanism.

In 2009, the 14 year old Brazilian Fábio Jardim became world champion by winning the contest *World Cyber Games (WCG)*. In the same year, Fábio posted videos on the Internet playing the music TTFAF on Expert level reaching 880,862 points and 97% accuracy [19]. Also in 2009, Danny Johnson broke the world record the highest score made on the TTFAF music, 985,206 points and 100% accuracy [20], being included in the Guinness Book of World Records.

VI. Conclusion

This paper presents a didactic robot dedicated to play the game Guitar Hero 3 on a PlayStation 2 console. The robot Grybot uses an FPGA chip to process in real-time composite video signal provided by console. Musical notes are detected by monitoring pixel colors. Synchronized signal commands are processed and sent to actuation system. These signals control electromechanical solenoids that press buttons of a DualShock2 controller. The Grybot successfully is able to play all songs from the repertoire of the game, at any difficulty level, reaching at least 95 % accuracy. Comparisons with other similar robots were made demonstrating the good performance achieved by the proposed robot. The VGA signal processing requires a low computational complexity compared to classical image processing techniques using cameras. It allows implementation in a 50 MHz FPGA chip, with no need of using computers nor specialized processors. The mechanical actioning using solenoid, similar to that performed by human players, has shown adequate. Robots that directly command the controller circuitry have a shorter response time, however it is a very intrusive approach.

In order to enhance the robot performance and to add new features to the system, we propose as future work the improvement of the detection of star-shaped notes and subsequent activation of *Star Power* to increase the reached score; the implementation of multiplayer mode allowing matches against a human player; and design and construction of a faster and more robust mechanical drive system.

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